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Python Test - 3

① for loop

1] write a program to print the multiplication table of a given number using for loop.

→ for loop :

```
num = int(input("Enter a number"))  
for i in range(1, 11):  
    print(num, "x", i, "=", num*i)
```

O/P : Enter a number = 4

①

2] write a program to print all prime numbers between 1 & 100 using for loop.

for num in range(2, 101):

prime = True

for i in range(2, num):

if num % i == 0:

prime = False

break

if prime:

print(num)

①

3] write a program to find the sum of all even numbers from 1 to n using for loop.

→ for n = int(input("Enter a number:"))

sum = 0

for i in range(2, n+1, 2):

sum = sum + i

print("sum of even numbers = ", sum)

①



4. OOPS

1] what is object-oriented programming (OOP)?

OOPS is a object-oriented programming paradigm that sof object software design occurs rather than object.

①

2] explain the difference between class & object with an example

① class :- A class is a Blueprint or template form a creating object

e.g. class student:

pass

①

② object :- an object is a instance of class

e.g. class car:

pass

c₁ = car(c)

3] what are the four pillars of oops? explain each

① Inheritance :- ① single inheritance

②

② multiple inheritance

③

3) multilevel inheritance

④

4) Hierarchical inheritance

4) Hybrid inheritance

② Encapsulation - wrapping data

① Public - variable

② protected - variable

3) private - variable



- 3) polymorphism -
- ① method overloading ✓
 - 2) method overriding ✓
 - 3) Polymorphism using inheritance ✓
 - 4) polymorphism us duck typing ✓
 - 5) Data abstraction operator overloading ✓
- 4] Data abstraction - module ABC ✓

4] what is inheritance? explain its type
— inheritance is the ops concept a child

- ① Single inheritance :- one parent → one child
- 2) multiple inheritance :- many parent → one child
- 3) multilevel inheritance :- inherits the level
- 4) Hierarchical inheritance - one parent-many children
- 5) Hybrid inheritance - combination multiple type

5] what is polymorphism? explain method overriding & over method overloading.

— the same method name behaves differently depend on a object ✓

- ① method overloading - does python does not support method overloading
- 2) method overriding - depend of parent class

6] Encapsulation what is Encapsulation? what are its advantages

— Encapsulation is a wrapping data and method into single class restrict restricting data



Advantage -

- ① Data protection
- ② impro security
- ③ controlled access

9.) explain the use of the super() function in python.
 → super() use to parent class method or constructor.

8.) what is abstract difference between abstraction and Encapsulation

① ~~abst~~ Encapsulation -

- ① Data hiding
- ② Data protected

② abstraction -

- ① implementation hiding
- ② Data imp provide

③ function :-

④ what is a function in python & explain its advantage
 — Function is a block of statement

* A function is a block of code that perform a specific task.

* Advantages -

- ① avoid Code Repetation

② make code readable

③ Reusability.

④ debugging.

5.) difference between user-defined function & built in func with e.g.

→ ① built-in function -

- ① already provided

② e.g. print(), len()

② user-defined function -

- ① created by def

② e.g. add()